

EA Credit: Grid Harmonization

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To increase participation in demand response technologies and programs that make energy generation and distribution systems more affordable and more efficient, increase grid reliability, and reduce greenhouse gas emissions.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Design building and equipment for participation in demand response programs through load shedding or shifting. On-site electricity generation does not meet the intent of this credit.

Case 1. Demand Response Program Available and Participation (2 points)

- ▶ Participate in an existing demand response (DR) program and complete the following activities. Design a system with the capability for real-time, fully-automated DR based on external initiation by a DR Program Provider. Semi-automated DR may be utilized in practice.
- ▶ Enroll in a minimum one-year DR participation amount contractual commitment with a qualified DR program provider, with the intention of multiyear renewal, for at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance. The on-peak demand may vary based on the utility climate and pricing structures.
- ▶ Develop a comprehensive plan for meeting the contractual commitment during a Demand Response event.
- ▶ Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- ▶ Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- ▶ Initiate at least one full test of the DR plan.

OR

Case 2. Demand Response Capable Building (1 point)

Have infrastructure in place to take advantage of future demand response programs or dynamic, real-time pricing programs and complete the following activities:

- ▶ Install interval recording meters and have equipment capable of accepting an external signal.
- ▶ Develop a comprehensive plan for shedding at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance.
- ▶ Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.

- ▶ Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- ▶ Contact local utility representatives to discuss participation in future DR programs.

AND/OR

Case 3. Load Flexibility and Management Strategies (1-2 points)

Analyze the building's annual load shape and peak load based as calculated for EA prerequisite Minimum Energy Performance. Review the regional grid load profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's (EPA) AVOIDed Emissions and geneRation Tool (AVERT) provides regional grid emissions data; local utilities may also provide this data.

Coordinate review of building load shape and peak load with review of the regional grid profile to identify the best value load management strategies that the building can provide.

Implement one or more of the load flexibility and management strategies described below. All projects must install interval recording meters and have equipment capable of accepting an external signal.

Load Flexibility and Management Strategies:

- ▶ Peak Load Optimization: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand referenced to the ASHRAE 90.1-2016 compliant case (1 point)
- ▶ On-site thermal and/or electricity storage: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)

Include installed technology in the scope of work for the commissioning authority. Include load flexibility and management strategies and installed technologies in the building systems manual, or include in the current facilities requirements and operations maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.

Contact local utility representatives to discuss participation in future DR programs and to inform utility of building load flexibility and management strategies.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

As the number of distributed energy resources, or grid-connected power generation from individual buildings, increases, the utility must integrate these resources while continuing to manage overall grid capacity and generation. Too much distributed generation that is not integrated into the energy system can impair grid operation. Buildings have the opportunity to support effective grid management by designing on-site electricity systems that are integrated components of the energy system, serving as resources for the grid rather than disruptors.

In addition to participation in a utility's demand response program, projects can install storage technologies and implement operational strategies that support effective grid management and increase grid reliability. Eligible technologies may include battery storage, flywheel energy storage, or thermal energy storage; load management strategies may include load shifting or flexible operating scenarios.

Beta Update

Updates intend to address projects where no demand response program is available from the local utility and recognize new distributed energy resources technologies and grid harmonization strategies.

Step-by-Step Guidance

Refer to Steps 1 – 5 in the LEED v4 reference guide, with the following addition:

Step 6. Include demand response in commissioning and in building systems manual
Coordinate with the CxA to include a review of the DR plan in the commissioning of the building's system test procedures, to verify the ability to handle an externally initiated demand response event. The Cx plan must include at least one performance test of the full DR plan to verify that all equipment responds as planned and that all responsible parties understand their roles.

Include demand response program and any installed technologies in the building systems manual.

Case 1. Demand Response Program Available

Refer to the LEED v4 reference guide EAc Demand Response, with the following addition:

Step 2. Initiate at least one full test of the DR plan.
Execute at least one performance test of the full DR plan as defined in the Cx plan. All relevant DR operations team members must participate, and address and mitigate any issues identified as a result of the performance test.

Case 2. Demand Response Capable Building

Refer to the LEED v4 reference guide, Case 2. Demand Response Program Not Available, with the following change:

Under the LEED v4.1 update, project eligibility for Case 2 is no longer restricted to applications where a demand response program is not available. However, the project team must still contact the utility service provider to discuss potential participation in a current or future DR program.

Case 3. Load Flexibility and Management Strategies

Step 1. Analyze building load shape

Analyze building load shape as calculated for EA prerequisite Minimum Energy Performance; compare building load shape with the regional grid profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's AVOIDed Emissions and geneRation Tool (AVERT) provides hourly regional grid emissions data or contact the local utility to request this data.

Step 2. Identify highest-value strategies

Based on the analysis of building load shape, peak load and regional grid peak load or peak carbon emissions, identify storage technologies or load management strategies that the building can implement at the lowest cost while providing the greatest value to the grid.

Step 3. Implement strategies

Implement one or both of the strategies identified during step 2.

Eligible strategies include peak load optimization achieved by reducing on-peak load by at least 10% compared to peak electrical demand, or installation of on-site thermal energy storage for heating and cooling and/or electricity storage capable of reducing on-peak load by at least 10% as compared to peak electrical demand. On-site storage enables the building to store energy and use it during peak demand times, increasing annual energy savings and reducing strain on the grid.

Step 4. Include load flexibility and management system in commissioning and in building systems manual

Coordinate with the CxA to include a review of the load management system and supporting technologies in the commissioning of the building's system test procedures. The Cx plan must include at least one performance test of the full system to verify that all equipment operates as planned and that all responsible parties understand their roles.

Include the system and any installed technologies in the building systems manual.

Step 5. Contact local utility

Contact the local utility or service provider to express interest in participation in a future demand response program, and to inform the utility of building load flexibility and management strategies.

Further Explanation

On-Peak Electricity Demand

On-peak electricity demand can be determined as one of the following for the purposes of credit compliance:

1. Building peak electricity demand
2. Building demand load coinciding with regional grid peak demand
3. Building demand load coinciding with regional grid peak carbon emissions

Grid Harmonization

Energy efficient and grid-interactive buildings save money and resources while supporting broader grid-scale decarbonization. Savings to building owners accrue from a combination of demand charge reductions, lower annual energy use, utility incentives, and increased grid resilience.

There are a number of methods by which projects can comply with the requirement in Case 2 and Case 3 to install equipment capable of responding to an external signal. External signals can be sent from a variety of sources such as the electric utility, the electricity Independent System Operator (ISO), or a third-party automation organization. The signal can be used to indicate specific events such as a critical price peaks, grid reliability events, high grid demand, or high grid GHG emissions factors; or a continuous signal reporting price, reliability, GHG emissions, or other factors. This equipment may be part of a building automation system, stand-alone equipment, or equipment that is directly integrated with the building system(s) used to achieve the credit, e.g. a packaged rooftop HVAC system.

Peak Load Optimization

To demonstrate that the peak load optimization strategy reduces on-peak load by at least 10% as compared to peak electrical demand, the baseline should be referenced to the ASHRAE 90.1-2016 compliant case. The "Performance Demand Index Target" and "Performance Demand Index" shall be calculated using the same methodology as the PCIT and PCI calculated per ASHRAE 90.1-2016 Appendix G:

$$\text{Performance Demand Index Target} = [\text{Baseline Building Unregulated Electric Demand} + (\text{BPF} \times \text{Baseline Building Regulated Electric Demand})] / \text{Baseline Building Demand}$$

For example, if the ASHRAE 90.1-2016 Building Performance Factor is 0.58, the Baseline regulated demand is 75 kW, and the Baseline unregulated demand is 25 kW for an Office building in climate zone 4A, the Performance Demand Index Target would be calculated as:

$$\text{Performance Demand Index Target} = [25 + (0.58 \times 75)] / (25 + 75) = 0.69$$

The proposed demand would need to show a 10% improvement beyond this Performance Demand Target. Therefore, for this application:

$$\text{Proposed demand} \leq (0.69 \times 100) \times 90\% = 62.1 \text{ kW}$$

In cases where the proposed case incorporates electrification of space heating and/or service water heating compared to the baseline case, project teams have two options for demonstrating peak load optimization:

- ▶ If able to demonstrate 10% on-peak load reduction with fuel switching, the project team may complete the calculation without applying corrections to the baseline.
- ▶ If unable to demonstrate 10% on-peak load reduction with fuel switching, the project team may increase the baseline building demand by the baseline case fossil fuel heating and service water heating demand divided by 2.0.

On-site Thermal and/or Electricity Storage

The reference case for demonstrating peak demand reduction via the On-site thermal and/or electricity storage strategy is the Proposed design from the Optimize Energy Performance energy model without the thermal and/or electricity storage strategy in place.

Single Strategy, Multiple Points

In some cases, a single design or operational strategy may meet the criteria for more than one strategy. For example, a set of demand reduction strategies used to achieve Case 2. Demand Response Capable Building may also meet the criteria for Case 3. Load Flexibility and Management Strategies for Peak Load Optimization.

Project Type Variations

District Energy Systems

When district energy is modeled for credit in EA credit Optimize Energy Performance, the district energy system (DES) electric demand modeled there must be included in the total demand reported for this credit. When the DES system is included in the EA credit Optimize Energy Performance model, grid harmonization strategies applied to the DES system may be used to document achievement at the building level.

The interval recording meter and building automation system communications may be located in the DES – and not in the building – if strategies in the DES are used to demonstrate full achievement of the credit requirements at the building level.

Required Documentation

Documentation	Case 1	Case 2	Case 3
Proof of enrollment in DR program	X		
Evidence of ability to shed 10% of on-peak demand	X	X	X
Confirmation that system is capable of receiving and acting on external signal	X	X	X
Action plan for meeting reduction requirement during event	X	X	X
Inclusion of DR in CxA systems testing plan	X	X	X
Inclusion of DR and/or grid harmonization technologies in building systems manual	X	X	X
Documentation of one full test of the DR plan	X		
Narrative or report that includes: summary of building annual load shape and regional grid profile analysis; description of building load flexibility and/or management strategies implemented			X
Documentation of technologies serving the project, as applicable			X

Connection to Ongoing Building Performance

- Participation in a demand response program enables projects to support efficient energy generation and distribution systems, increase grid reliability, and reduce greenhouse gas emissions. For projects where no demand response program is available, implementing load flexibility and management strategies can help to achieve the same outcomes and support effective electrical grid management.

Exemplary Performance

Achieve Case 1 with the capability for at least a 20% peak demand savings, or achieve at least three points under Case 1 and Case 3, or achieve at least three points under Case 2 and Case 3.